



Diagnosis and pathogenesis of dystonia: clinical heterogeneity, shared mechanisms, and neurodevelopmental origins

Emmanuel Roze, Verity M McClelland, Lara M Lange, Hyder A Jinnah, Marie Vidailhet, Antonio Pisani

The diagnosis of dystonia is often delayed because of its many different clinical manifestations and causes. Updates to the definition and classification of dystonia can assist clinicians in improving its recognition and timely diagnosis. Dystonia is characterised by sustained or intermittent abnormal movements, postures, or both. Patterned phenomenology, alleviating manoeuvres, worsening by voluntary movement, and overflow to adjacent muscles are all clinical clues of the condition. Additional non-motor manifestations are part of the clinical spectrum. In parallel to the broad heterogeneity of clinical manifestations among different types of dystonia, the biological mechanisms underlying the condition are also heterogeneous. Among these mechanisms, an alteration of the sensorimotor circuits during crucial windows of development might have a role in many forms of dystonia. Advances in the understanding of its pathogenesis indicate that targeting shared mechanisms could provide a treatment approach for multiple types of dystonia, but also suggest the possibility of personalised therapeutic approaches.

Introduction

The diagnosis of dystonia remains challenging because of poor recognition of its different clinical manifestations and causes. Additionally, there are no widely applicable diagnostic tests for dystonia. Of note, the availability of genetic tests and other diagnostic tools differs between countries, which can hamper the identification of the aetiology of dystonia in some settings. Over the past decade, refinement of the clinical characterisation and classification of various dystonic disorders, along with advances in understanding the anatomical and physiological mechanisms underlying dystonia, have transformed the clinical approach to this condition.¹

In this Review, we cover advances in the diagnosis of dystonia, in the identification of its underlying cause, and in the classification of patients in clinical or research settings. We also synthesise recent advances in understanding pathogenesis, focusing on its neurodevelopmental origins. Within this context, we highlight relevant mechanisms on the basis of their novelty and potential for future preventive or curative treatment.

Diagnosis

Dystonia is commonly encountered in movement disorder clinics and the term refers to a well-defined clinical phenomenology. Diagnosis is based primarily on motor signs, characterised by sustained or intermittent abnormal movements, postures, or both. Dystonic movements and postures are typically patterned and repetitive, and can be tremulous or jerky.¹ Dystonia is often initiated or worsened by voluntary action, with a so-called overflow of muscle contractions to an adjacent or distant body segment.^{1,2} The existence of an alleviating manoeuvre (also referred to as *geste antagoniste* or sensory trick) is a common feature. This specific voluntary movement or light touch, usually applied to a particular body part, can be used by patients to achieve transient relief from dystonic contraction.³ Patients often manifest

reduced dystonic muscle activity when relaxed or during sleep, which can persist transiently on waking.⁴ Of note, an improvement of symptoms in response to alcohol can occasionally be observed in individuals with subtypes of dystonia, especially in patients with early-onset multifocal or generalised dystonia. The sequential steps of the clinical assessment of individuals with dystonia followed in our clinical practice are described in the appendix (p 3).

Dystonia is often associated with tremor, myoclonus, or both. These abnormal movements can be observed in the same body areas affected or in those spared by dystonia. The prevalence of tremor is positively correlated with the severity and duration of dystonia.⁵ According to expert recommendations updated in 2025, clinicians should describe this association, noting whether both tremor and dystonia occur in the same or different body regions.⁶ The recommended terms, dystonia tremor syndrome or dystonia with tremor, describe the coexistence of dystonia and (rhythmic) tremor, whereas jerky dystonia characterises a type of dystonia with arrhythmic jerks, typically corresponding to rapid and repetitive dystonic movements.⁶

Although dystonia is defined as a motor disorder, associated non-motor manifestations can include pain, psychiatric problems (mainly anxiety and depression), sleep disorders, fatigue, and sometimes subtle cognitive difficulties.^{7,8} Additionally, some patients can experience stigma, social withdrawal, isolation, loss of self-confidence and self-esteem, with poor body image, and impaired social cognition with emotional and other consequences.⁹ Non-motor problems, especially anxiety and depression, are strong independent determinants of quality of life in patients with dystonia.¹⁰ It is still unclear whether non-motor manifestations are a primary manifestation or a consequence of motor problems. However, some preliminary data suggest that psychiatric features are a primary component.^{11,12} Therefore, the

Lancet Neurol 2026; 25: 591–601

Sorbonne University Paris Brain Institute, Inserm, CNRS, Paris, France (E Roze MD, M Vidailhet MD); AP-HP, Salpêtrière Hospital, Institute of Neurology, Paris, France (E Roze, M Vidailhet); Department of Clinical Neuroscience, Institute of Psychiatry, Psychology and Neuroscience, King's College London, London, UK (V M McClelland PhD); Department of Paediatric Neurosciences, Evelina London Children's Hospital, London, UK (V M McClelland); Institute of Neurogenetics, University of Lübeck, Lübeck, Germany (L M Lange MD); Department of Neurology, University Hospital Schleswig-Holstein, Lübeck, Germany (L M Lange); Department of Neurology, Department of Human Genetics, and Department of Paediatrics, Emory University School of Medicine, Atlanta, GA, USA (H A Jinnah MD); Department of Brain and Behavioural Sciences, University of Pavia, Pavia, Italy (A Pisani MD); IRCCS Mondino Foundation, Pavia, Italy (A Pisani)

Correspondence to: Prof Emmanuel Roze, AP-HP, Salpêtrière Hospital, Institute of Neurology, Paris 75013, France emmanuel.flamand-roze@aphp.fr

See Online for appendix

evaluation of non-motor manifestations should be added to the routine clinical assessment of individuals with dystonia. Future clinical research will need to design and validate dedicated tools for the accurate evaluation of non-motor manifestations, similar to those that have been developed for the evaluation of pain in these patients.¹³

Differential diagnosis

Clinical tools to guide clinicians in the diagnosis of dystonia have been proposed elsewhere and are beyond the scope of this Review.^{14,15} Many conditions can mimic dystonia, making the differential diagnosis challenging. Some clinical red flags can help to differentiate dystonia from its mimics (table). However, some of these signs (eg, abrupt onset or tonic postures) can also be rarely observed in patients with typical dystonia, such as in those with rapid-onset dystonia parkinsonism. Similarly, paroxysmal movements triggered by emotional factors can reflect a genetic paroxysmal disorder or a functional movement disorder.

Classification

Considering the progress made in the characterisation and diagnosis of dystonia over the past decade, an updated clinical toolbox for clinicians and researchers has emerged to facilitate the accurate clinical description and diagnosis of dystonia subtypes in clinical settings, as well as standardised data collection in research settings.¹

The many subtypes of dystonia are classified using two axes. Axis I focuses on phenomenology and recognition of the dystonia syndrome, including age at onset, family history, body distribution, temporal dimension (onset, course, and variability within 1 day), and associated features (figure 1). Axis II focuses on disease aetiology and mechanisms, including categorisation of the underlying nervous system pathology and distinction between genetic and acquired causes.

Focal dystonia is the most frequent form of adult-onset dystonia. In movement disorders clinics, most patients are referred with idiopathic, isolated, adult-onset focal or segmental dystonia. Focal dystonia involves only one body region (upper face, lower face, neck, larynx, upper limbs, trunk, or lower limbs).¹ Over the course of the disease, dystonia can spread to muscles beyond the initially affected region. In an international cohort study, such a spread was found in 58 (50%) of 116 patients with blepharospasm (to the oromandibular region or the neck), eight (17%) of 47 patients with hand dystonia, six (16%) of 38 patients with laryngeal dystonia, and 24 (8%) of 286 patients with cervical dystonia.¹⁶

Cervical dystonia is the most prevalent form of focal dystonia affecting the head and neck, with or without shoulder involvement.¹⁷ Although traditionally described by the main direction of the head and neck movement, a recent characterisation describes differing degrees of head or neck involvement.^{18,19} Accurate delineation of

	Type of dystonia mimicked	Motor signs	Signs and symptoms	Additional clinical clues
Functional neurological disorder	Any type of dystonia (eg, facial or limb)	Variability of intermittent or permanent contractions	• Ipsilateral downward and lateral jaw pulling • Foot plantar flexion and inversion; first toe extension and flexion of other toes • Wrist and finger flexion, sometimes with a clenched fist • Inconsistent features and patterns during or between examinations	• Abrupt or progressive onset • Fixed posturing at onset • Precipitating factor (in 50–80% of cases) • Fluctuations and distractibility • Reduction or abolition with distraction; symptoms increasingly triggered by focused attention
Benign infantile torticollis	Cervical dystonia	Fixed lateral inclination of the neck and permanent mild neck rotation	• Sternocleidomastoid muscle atrophy, muscle shortening, and fibrosis	• Onset shortly after birth or in the neonatal period • Stable over time • Not disabling
Grisel syndrome	Cervical dystonia	Abnormal posture of the neck or pseudo-neck dystonia	• Resistant to passive neck movements • Fixed postures • Equally present both in rest and action • Rotatory atlantoaxial subluxation • Inflammatory or infectious condition in the head or neck	• Atraumatic onset • Acute onset of painful torticollis • Primarily in children
Stiff person syndrome* and PERM†	Cervical and truncal dystonia	Painful and stiff neck, trunk, or both	• Stiff posture and muscles (no abnormal movement) • Painful neck and shoulder muscles; continuous muscle contraction	• Painful • Progressive • Non-motor symptoms
Anti-IgLON5 disease	Cervical dystonia	Head drop with sustained and recurrent bending of the neck	• Sleep disorders • Cognitive disorders • Hallucinations	• Progressive evolution

PERM=progressive encephalomyelitis with rigidity and myoclonus. *Anti-glutamic acid decarboxylase antibodies. †Anti-glycine receptor antibodies.

Table: Signs and symptoms of some conditions that can mimic dystonia

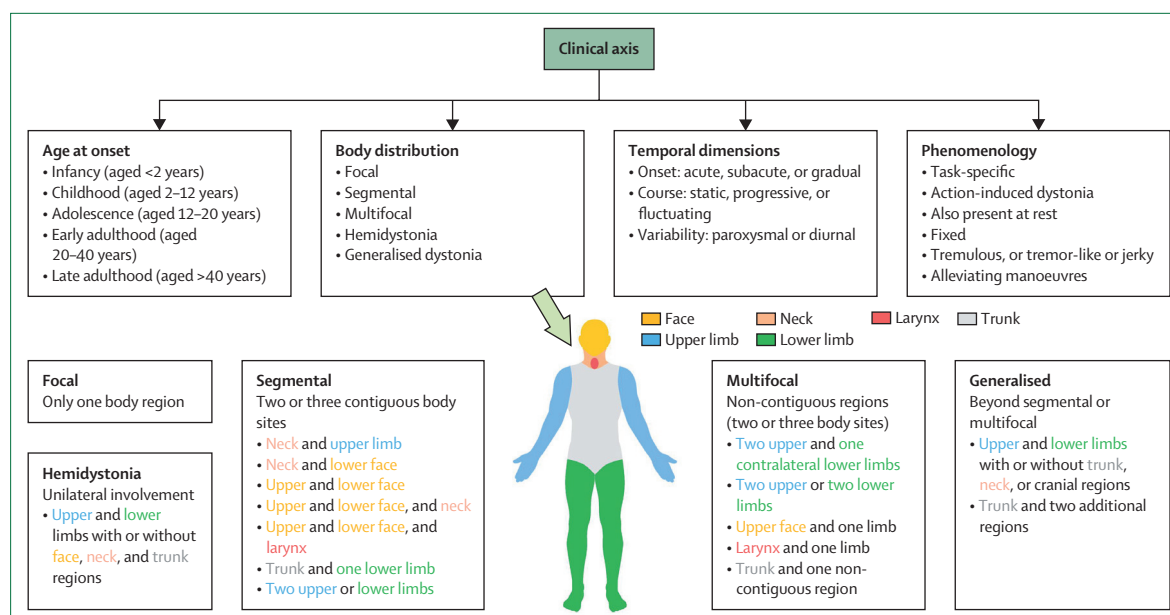


Figure 1: Clinical characteristics of dystonia

Axis 1 of the dystonia classification comprises six descriptive components. The four main components are age at onset, body distribution, temporal dimensions (onset and evolution), and phenomenology (especially in relation to voluntary movement). The two remaining components (not shown) are family history (sporadic, familial, or unknown) and associated features (isolated dystonia or dystonia associated with other movement disorders, neurological features, or systemic features). The schematic diagram of body regions affected by dystonia depict representative examples of focal, segmental, multifocal, and generalised dystonia.

the isolated or combined alterations of neck and head position can help to improve muscle targeting for local treatment with botulinum toxin injections. These alterations of head and neck position include rotation (torticollis or torticaput), flexion (antecollis or antercaput), extension (retrocollis or retrocaput), lateral tilt (laterocollis or laterocaput), or shift [lateral [neck shifted towards one shoulder] or anterior or posterior shift [neck shifted forwards or backwards]].^{18,19}

Dystonia can also involve the craniofacial region, including the muscles around the eyes (blepharospasm) or the oromandibular region. A disabling feature that can accompany facial dystonia is pretarsal blepharospasm (also called apraxia of eyelid opening), characterised by difficulties in voluntarily opening the eyes despite compensatory manoeuvres, such as sustained frontalis contraction.²⁰ Oromandibular dystonia consistently induces eating difficulties (jaw closing, jaw opening, or a combination of both). Lingual involvement is less often observed.²¹ Laryngeal dystonia mainly manifests with difficulties in vocalisation (adductor-type with strained strangled voice or abductor-type with whispering voice).

Task-specific dystonia is another form of focal dystonia, in which dystonic movements and postures are triggered exclusively by the performance of a specific learned activity. Musician's dystonia is the prototypical task-specific dystonia, associated with repetition of over-learned, complex, and skilled movements that push the limits of human motor

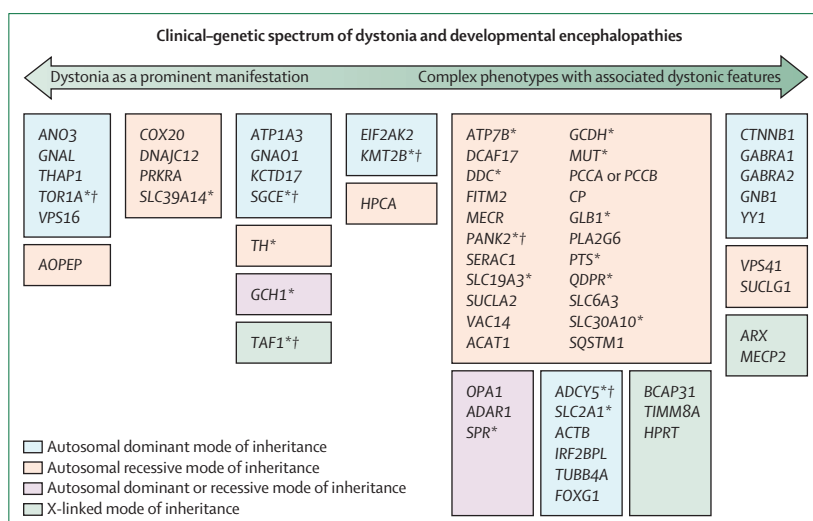


Figure 2: The clinical-genetic spectrum of dystonia and developmental encephalopathies

The clinical-genetic spectrum of dystonia ranges from conditions in which dystonia is a prominent manifestation to complex phenotypes with associated dystonic features. Each gene is placed according to the clinical presentation that best characterises most patients. Genes are colour-coded to indicate their mode of inheritance. Several genes could have been placed at multiple positions along the spectrum due to their broad phenotypic heterogeneity. *Conditions that are potentially treatable, such as dopa-responsive dystonia or dystonia caused by treatable neurometabolic disorders. †Conditions known to respond well to deep brain stimulation.

control,²² likely involving a mismatch between task requirements and the plastic capacity of the motor system. Task-specific dystonias are also described in other occupations or professions involving skilled movements,^{22,23} such as in sports people (golf, table

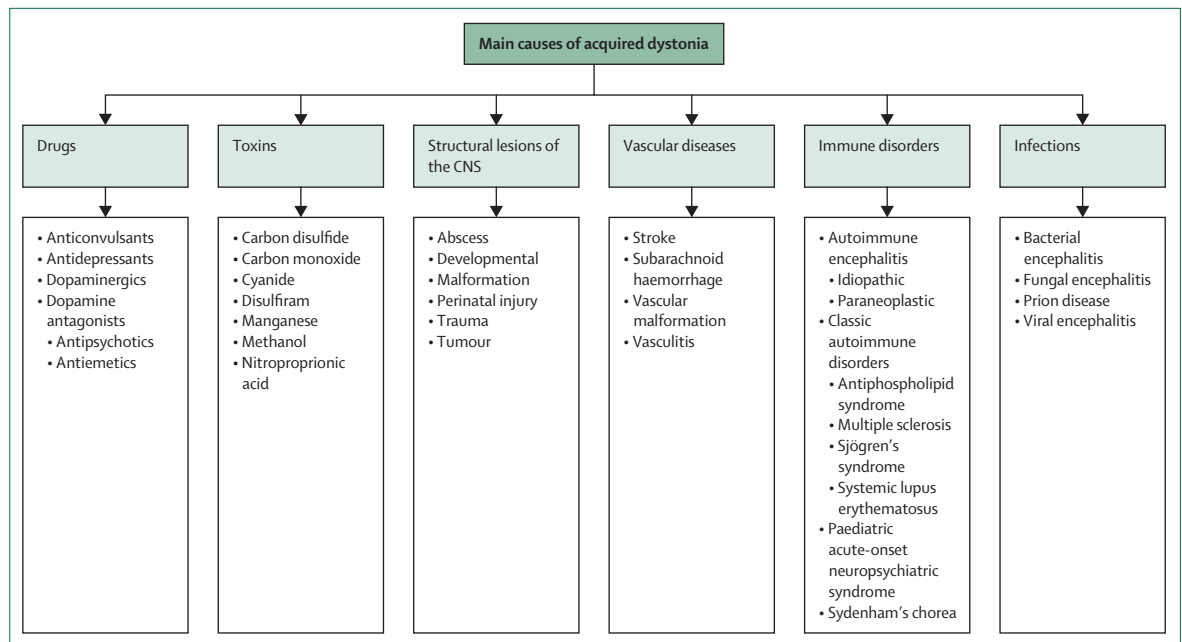


Figure 3: Main causes of acquired dystonia

The main causes of acquired dystonia can be divided into six groups: drugs, toxins, structural lesions of the CNS, vascular diseases, immune disorders, and infections.

tennis, ball games, running, gymnastics, and electronic sports). Writer's cramp is another example of task-specific dystonia, involving the development of a variable combination of abnormal postures of the fingers, hands, and wrist that disrupt writing performance and legibility. Dystonic postures can also develop over time for other activities, leading to loss of dexterity, and they can eventually persist at rest.

Pathogenesis

In parallel with the broad heterogeneity of clinical manifestations, the biological mechanisms underlying dystonia are also highly heterogeneous.²⁴ Although most adult-onset cases are idiopathic, other types of dystonia are genetic, with the potential involvement of hundreds of genes (figure 2).^{24–27} Some of these genes have been associated with neurodegenerative processes. Dystonia might also be acquired, and many triggers have been described (figure 3). Illustrative cases with genetic or acquired forms of dystonia are briefly described in panel 1, videos 1–6, and the appendix (pp 1–2). A proportion of individuals with dystonia have overt focal or diffuse brain lesions, but most patients show no evidence of any structural lesion.²⁸ Despite the heterogeneity of dystonia, there has been great interest in delineating shared mechanisms. These mechanisms might occur at different levels, including molecular, anatomical, or physiological. For example, multiple genes might converge on a molecular mechanism that causes dystonia. Alternatively, dystonia might emerge from the dysfunction of shared anatomical pathways or physiological processes. Importantly,

neurodevelopmental processes might be relevant in various forms of dystonia (including those with adult-onset), irrespective of aetiology. We focus specially on these neurodevelopmental aspects and their relevance, given that other mechanisms have been reviewed elsewhere.^{24,29,30}

Anatomical mechanisms

Dystonia is thought to arise from the dysfunction of an anatomical network that includes the basal ganglia, cerebellum, thalamus, and sensorimotor cortex.^{28,31} Exactly how the network is affected in the different types of dystonia is not fully understood. Dystonia might emerge following network dysfunction of a single region, simultaneous dysfunction of two or more regions, or abnormal inter-regional connections.²⁸ The concept of subgrouping different types of dystonia according to shared underlying anatomy is most apparent for dystonias caused by focal lesions.^{28,32} A 2022 study suggested that focal lesions of the basal ganglia are most often associated with limb dystonia, whereas lesions of the cerebellum and its brainstem connections are more often associated with cervical dystonia.³³ A shared anatomy might also apply to other biological mechanisms, even without obvious structural lesions. For example, overlapping patterns of regional glucose hypermetabolism or hypometabolism are seen across some dystonia aetiologies.³⁴ Some molecular mechanisms might trigger dystonia by affecting processes in the basal ganglia, whereas other mechanisms might first affect other regions, but without causing any obvious anatomical change.

See Online for videos 1–6

Physiological mechanisms

Dystonias can also be subdivided according to shared physiological mechanisms.³⁵ The main pathophysiological mechanisms involved in dystonia are an imbalance between inhibitory and excitatory neurotransmission in the CNS, abnormalities in sensorimotor integration, and maladaptive plasticity. Maladaptive plasticity might have a particularly important role in task-specific dystonia, like musician's dystonia, whereby repetitive practice triggers the development of abnormal movements. Maladaptive plasticity might be less relevant in other types of dystonia triggered by a pathogenic gene variant or an acquired focal lesion, in which other physiological mechanisms seem to be more important. However, when considered in the context of neurodevelopmental plasticity within the sensorimotor network, shared pathophysiological processes might still be pertinent.

Shared physiological pathways are often better characterised in experimental models of dystonia. Multiple genes can contribute to pathogenesis through the dysregulation of synaptic activity and plasticity in specific brain regions, such as the basal ganglia. The loss of striatal long-term depression, coupled with enhanced long-term potentiation, seems to be a common feature in multiple rodent models of dystonia.³⁶ In these studies, dopamine pathways are frequently implicated, either directly (eg, through dopamine receptor dysfunction as in *TOR1A*, *THAP1*, and *GNAL* mutations) or indirectly.^{37,38}

Although multiple experimental models of dystonia have identified common physiological changes in the basal ganglia, others have shown converging physiological abnormalities in the cerebellum. One particular physiological phenomenon identified in multiple unrelated animal models involves a change in the activity of Purkinje neurons from a normal pattern of relatively regular simple spikes, occasionally interrupted by complex spikes, to an abnormal pattern with frequent burst firing.³⁹ The current hypothesis is that dystonia is a network disorder, in which dysfunction of a given brain region causes adaptations in others. The clear demonstration of a link between altered activity in the cerebellum and motor features in models of dystonia strongly supports this hypothesis.³⁹

Molecular and cellular mechanisms

Multiple molecular mechanisms have been recognised, among which those converging on dopamine neurotransmission are the most extensively studied and the most relevant for clinical practice (figure 4). Dopa-responsive dystonia is most often caused by pathogenic variants in the *GCH1* gene, which encode a protein that is crucial in dopamine synthesis.²⁴ Numerous other proteins that affect dopamine synthesis (encoded by *SPR*, *PRPS*, *TH*, and *DDC*), dopamine transport (*SLC6A3* and *SLC18A2*), or postsynaptic signal transduction downstream of dopamine receptors (*GNAL* and *GNAO1*) can also cause dystonia, often in

Panel 1: Illustrative case studies of dystonia

Dopa-responsive dystonia

A female patient aged 12 years developed a bilateral action tremor of the upper limbs, that was associated with cervical dystonia at age 15 years. Her condition then evolved into generalised dystonia and bradykinesia. She was initially diagnosed with cerebral palsy in another institution and by age 16 years she was unable to walk and needed a wheelchair. The diagnosis of dopa-responsive dystonia was made at age 22 years, based on the dramatic clinical improvement observed 1 week after the initiation of levodopa (video 1A, B). The diagnosis was confirmed by genetic analysis (our patient was a monoallelic carrier of a R216X truncating variant in *GCH1*). This case illustrates the importance of considering levodopa administration in any patient with unexplained generalised dystonia, especially when onset is in childhood or adolescence.

Dystonia as the main manifestation of Wilson's disease

A female patient aged 16 years gradually developed right upper limb dystonia, dysarthria, and depression. She had a bilateral Kayser–Fleischer ring on examination of the eyes and, on MRI, her scan showed a bilateral hypersignal of the striatum and midbrain. We made a diagnosis of Wilson's disease based on low ceruloplasminaemia (0.08 mmol/L; normal range >0.16 mmol/L), increased copper in urine (7.4 mmol/L; normal range <0.4 mmol/L), and increased relative exchangeable copper (42%; normal range <15%). Despite treatment with copper chelation, her condition gradually worsened towards generalised dystonia, requiring liver transplantation at age 18 years (video 2).

Tardive dystonia

A female patient aged 42 years presented to our institution with an acute psychotic episode and was treated with risperidone and alimemazine, with favourable outcome. 1 year after the psychotic episode, risperidone was stopped and she developed generalised mobile dystonia (video 3). A comprehensive diagnostic investigation was otherwise unremarkable, and the diagnosis of tardive dystonia was made.

combination with other neurological manifestations. Dystonia also occurs in genetic conditions associated with degeneration of dopaminergic neurons (involving variants in *LRRK2*, *PRKN*, *PINK1*, *POLG*, *SNCA*, and *NR4A2*). Some genes linked to dystonia disrupt dopamine signalling (*HPRT1* and *TOR1A*), although the exact mechanisms are unknown.

Non-genetic mechanisms that affect dopamine neurotransmission might also cause dystonia. For example, tardive dystonia can emerge following chronic treatment with dopamine receptor antagonists.⁴⁰ Dystonia is a frequent feature of levodopa-induced dyskinesias in people with Parkinson's disease. In individuals with acquired brain lesions, dystonia correlates with loss of dopamine innervation in basal ganglia. Furthermore, PET studies indicate abnormalities of dopamine signalling in adult-onset idiopathic focal dystonias, such as writer's cramp or spasmodic dysphonia.

The exact mechanisms by which abnormalities of dopamine neurotransmission produce dystonia remain to be determined. In some cases, dystonia seems to result from presynaptic dopamine deficiency, given that dystonia can be treated with levodopa, as in patients with dopa-responsive dystonia or Parkinson's disease. In other cases, alterations in postsynaptic dopamine receptor signalling is a probable cause, as in people

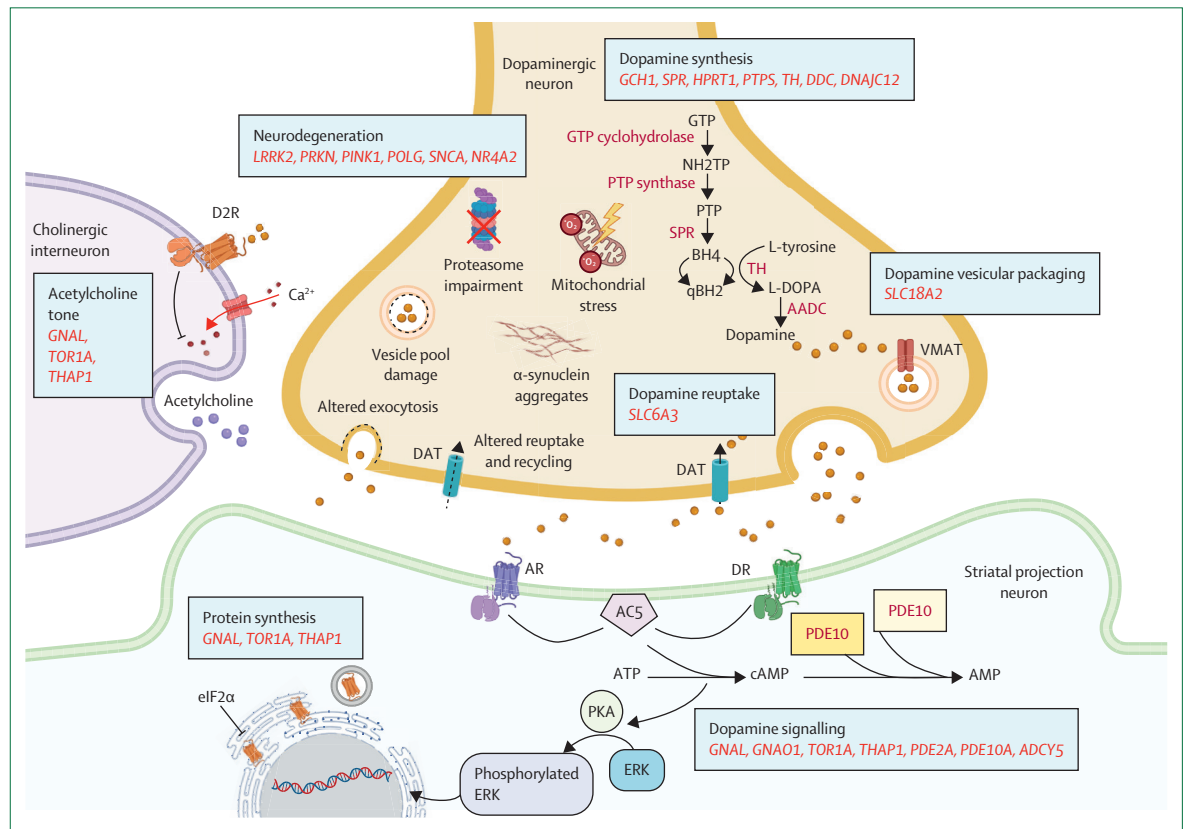


Figure 4: Altered striatal dopamine neurotransmission in dystonia

A schematic representation of dystonia-related pathways involved in striatal dopaminergic transmission at presynaptic and postsynaptic levels. In dopaminergic neurons, dopamine synthesis can be affected at multiple levels, either directly involving enzymes involved in its synthesis (ie, variants in TH, DDC, and HPRT1) or indirectly (GCH1, SPR, PTPS, and DNAJC12). Likewise, multiple genes linked to neurodegeneration of dopaminergic neurons (LRRK2, PARKIN, PINK1, POLG, SNCA, and NR4A2) can be involved in the alteration of dopamine transmission. Both presynaptic dopamine release in the synaptic cleft and reuptake mechanisms can be also affected, involving vesicular monoamine transporter VMAT2 (SLC18A2) and dopamine transporter DAT (SLC6A3). In striatal projection neurons, several genes are involved in the expression of proteins for post-receptor machinery of dopamine and adenosine receptors, including G protein-dependent pathways (GNAL and GNAO1) and cAMP-related signalling cascades (ADCY5, PDE2A, and PDE10A). The dystonia-linked genes, TOR1A and THAP1, also have a role in the stress response, acting on eIF2 α and secretory pathways in the endoplasmic reticulum, and altering dopamine signalling. Together with GNAL, TOR1A and THAP1 affect gene transcription. In striatal cholinergic interneurons, acetylcholine tone can be impaired by variants in dystonia genes (GNAL, TOR1A, and THAP1) through dopamine receptor modulation. AMP=adenosine monophosphate. ATP=adenosine triphosphate. BH4=tetrahydrobiopterin. cAMP=cyclic adenosine monophosphate. GTP=guanosine triphosphate. NH2TP=dihydroneopterin triphosphate. qBH2=q-dihydrobiopterin. Figure created with BioRender.com.

carrying variants in the *GNAL*, *GNAO1*, or *ADCY5* genes (figure 4). In other cases, the mechanism remains uncertain, given that therapies aimed at altering dopamine neurotransmission have no effect (eg, in adult-onset focal dystonia or dystonia with *HPRT1* or *TOR1A* variants).

Although dopamine signalling has a role in various unrelated forms of dystonia, it might not have a central role in all dystonias, and other molecular and cellular mechanisms have been recognised (figure 4). Some of the most frequently highlighted include brain accumulation of metals (iron, copper, manganese, or calcium), abnormal intracellular trafficking and function (due to impairments in nucleocytoplasmic transport, endoplasmic reticulum, or lysosomes), alteration of purine metabolism, mitochondrial dysfunction (both genetic and toxin-induced), and dysregulation of gene expression.^{24,26,41,42}

Therapeutic targets

Delineating shared mechanisms is of great interest in experimental therapeutics because targeting a shared mechanism could provide a treatment approach for multiple types of dystonia. A clear illustration of a shared therapeutic strategy based on physiology involves targeting excessive neuromuscular activity with botulinum toxin. This treatment is effective for all types of dystonia, regardless of the underlying mechanism.⁴³ However, the requirement for intramuscular injections limits its usefulness when dystonia affects many body regions.

Shared anatomical mechanisms are relevant for surgical treatment approaches. Deep brain stimulation of the pallidum is highly beneficial in some types of dystonia,⁴⁴ whereas targeting the thalamus is more effective for others.^{45–47} In some cases, neither target is effective, and further work is required to delineate optimal stimulation sites.⁴⁸

There has also been increasing interest in defining shared molecular targets. Dopamine pathways can be modulated in numerous ways. Other drugs, such as anticholinergics or benzodiazepines, seem to have broad efficacy across many types of dystonia in real-world settings. However, understanding the underlying mechanisms of therapeutic effects is of great importance for the development of alternatives with fewer side-effects and greater efficacy. Furthermore, a greater understanding of how shared mechanisms act during neurodevelopment supports the hypothesis that early intervention might be crucial for patients with some subtypes of dystonia.

Clinical targeting of shared mechanisms needs to be considered, alongside the important concept of a personalised therapeutic approach. For example, in some cases, the causative gene or variant can be a major determinant for selecting a dedicated, potentially life-changing treatment, such as dopaminergic treatment in dopa-responsive dystonia, ketogenic diet in dystonia due to GLUT1 deficiency,⁴⁹ caffeine in ADCY5-related dystonia,⁵⁰ copper chelation or zinc salts in dystonia due to Wilson's disease,⁵¹ or gene therapy in aromatic L-amino acid decarboxylase deficiency.⁵²

Dystonia as a neurodevelopmental disorder

Neurodevelopmental plasticity encompasses dynamic processes with a genetically programmed temporal profile, that lead to crucial sensitive windows of activity-dependent brain development in prenatal and postnatal periods.^{53,54} Herein, we consider dystonia within the context of neurodevelopmental plasticity and speculate about developmental factors that might be relevant to pathophysiology across aetiologies and age groups.

During late pregnancy and the first years of life, synaptic connections are determined intrinsically,⁵⁵ however, refinement of sensorimotor cortex organisation and circuitry is experience-dependent⁵⁶ and can be disrupted by perinatal injury.^{56,57} Abnormal sensory input during these crucial periods might result in permanent defects in neural circuit structure or function that the mature system is unable to compensate for.⁵⁸ In rat models, perinatal hypoxia–ischaemia and postnatal hindlimb movement restriction are associated with degradation of somatosensory cortical maps and increased cortical excitability.⁵⁹ These findings, which show parallels with the overlapping cortical receptive fields observed in focal hand dystonias,⁶⁰ are associated with the development of abnormal locomotor patterns.⁵⁹ Abnormal sensory-evoked potentials in neonates following asphyxiation are highly predictive of the development of cerebral palsy.⁶¹ Abnormal sensory-evoked potentials are also reported in some patients with acquired dystonia secondary to perinatal hypoxic–ischaemic encephalopathy.⁶² This observation does not demonstrate causation, but one hypothesis is that sensory pathway dysfunction is long-standing and affects

the activity-dependent modelling and refinement of sensorimotor circuits in the early postnatal period. Abnormalities in sensory-evoked potentials are not specific to dystonia and could simply reflect global neurological injury. However, in a cohort of children with dystonia of various causes, ten (25%) of 40 individuals with abnormal sensory-evoked potentials had typical cranial MRI imaging without evidence of widespread structural damage.⁶² In a study in rats, perinatal hypoxia provoked the development of specific electrophysiological measures of dystonia.⁶³ Furthermore, several studies in animals^{64–66} and humans⁶⁷ have shown changes in striatal cholinergic function across many dystonia aetiologies, thereby linking the pathogenesis of dystonia due to perinatal hypoxia to that of other causes of dystonia.

A variant in *TOR1A* is the most well characterised monogenic gene variant associated with dystonia. The Tor1A protein (also known as Torsin-1A) has important roles in synaptogenesis and synaptic plasticity, and Tor1A hypofunction is associated with abnormal synaptic transmission and plasticity in mouse models.⁵⁸ Suppression of Tor1A expression from embryogenesis onwards induces neuropathological and behavioural phenotypes reminiscent of *DYT-TOR1A*, whereas suppression in the fully mature state does not.⁶⁸ Additionally, restoration of Tor1A expression rescued the behavioural phenotype, but only in juvenile and not adult mice, suggesting that loss-of-function *TOR1A* variants exert their pathogenic effect within a crucial neurodevelopmental window.^{58,68} Interestingly, posterior expression of the putaminal acetylcholine transporter is decreased in young (aged ≤ 42 years), but not older (> 42 years), patients with *DYT-TOR1A*, providing human evidence for disruption to corticostriatal motor networks in an early developmental window.⁶⁷

Any abnormality that deprives brain networks of activity during crucial periods of neurodevelopment affects not only the structure and connectivity of sensorimotor circuits under development, but also their relative excitability. This excitability is mediated through homeostatic plasticity mechanisms,⁶⁹ which vary across brain regions and developmental stages.^{53,69} Thus, abnormalities in early circuit development, of either genetic or acquired origin, might be offset by enduring changes in synaptic function and excitatory–inhibitory balance. We speculate that, in some genetic dystonias, abnormal neuritogenesis during a crucial neurodevelopmental window is offset by synaptic changes. In turn, this process leads to a compensatory change in the plasticity response and its range, enhancing the effect of long-term potentiation-like mechanisms.⁵³ Emerging evidence in several monogenic dystonias suggests that molecular mechanisms drive circuit changes that affect plasticity processes.⁷⁰ In acquired dystonias due to perinatal brain injury, sensorimotor circuits are suddenly exposed to a profound alteration of activity in the corticobasal ganglia–thalamo–cortical loop, the

cortico–ponto–cerebello–thalamo–cortical loop, or both, adversely affecting the ongoing modelling and refinement of sensorimotor circuits. In this context, the relative but sudden deprivation of afferent input, even if transient, could lead to a change in the so-called set point around which these processes function,⁵³ reflecting a different form of maladaptive neuroplasticity. The development of dystonic manifestations following chronic, but not acute, striatal cholinergic interneuron excitation in a mouse model of cerebral palsy is concordant with a maladaptive plasticity response.⁶⁶

The principle of using crucial developmental windows to restore a phenotype through early intervention has

been investigated in a rat model of cerebral palsy. Early intensive rehabilitation in rats induced changes in the expression of synaptic function proteins in the brain and spinal cord, indicating a neuroplastic response, associated with the amelioration of abnormal locomotor patterns.⁷¹ A 2021 meta-analysis further supports targeted early intervention for infants with or at high risk of cerebral palsy (not specific to dystonia or dyskinesia).⁷² The potential for early intervention in dystonia will first require detailed delineation of underlying mechanisms within the context of neurodevelopment, based on consolidated evidence from across the life course. This approach must also account for the fact that some developmental abnormalities and their consequences might be reversible, whereas others might not.

Evidence from large-scale sequencing studies^{73–76} indicates a molecular intersection between dystonia and developmental encephalopathies. First, dystonic features can occur in patients with genetic alterations in classic genes linked to developmental encephalopathies. Second, variants in genes linked to isolated dystonia can cause broader phenotypes, including features of developmental encephalopathies. Third, genetic alterations in genes causing developmental encephalopathies and dystonia can induce dysfunction of similar pathways. Therefore, isolated dystonia might be seen as a trait reflecting one end of the phenotypical continuum of neurodevelopmental disorders, whereas developmental encephalopathies without comorbid dystonia mark the other end of this spectrum, with many intermediate phenotypes in between (figure 2). Although the overlap is most pronounced in childhood-onset forms, genetic alterations in developmental encephalopathy genes are increasingly recognised in adult-onset dystonia. Clinically, this continuum is exemplified by epilepsy-dyskinesia syndromes, a large and heterogeneous group of conditions defined by the co-occurrence of epilepsy and movement disorders. Around 90% of individuals with epilepsy-dyskinesia syndromes show neurodevelopmental features—most commonly developmental delay or intellectual disability—and, in these patients, dystonia is the most frequent movement disorder.⁷⁷ Similarly, across genetic epilepsies and related encephalopathies, dystonia and other hyperkinetic movements are increasingly recognised as integral features of the phenotype.⁷⁸

How can genetic alterations that induce aberrant neurodevelopment, sometimes within the same gene, cause either a plethora of phenotypes or the same phenotype? One answer might be variant-specific effects; truncating variants could influence neurodevelopmental processes to a different extent to missense variants, resulting in differential clinical phenotypes.^{79,80} Additionally, polygenic, epigenetic, and environmental factors might influence phenotypic diversity. Alternatively, downstream consequences of developmental brain dysfunction, rather than the genetic variant per se, might primarily influence the phenotype.

Panel 2: Knowledge gaps and research priorities

Knowledge gaps

Clinical diagnosis and heterogeneity

- Early identification of dystonia remains challenging due to broad clinical variability.
- The drivers of clinical heterogeneity are poorly understood.
- No general biomarker exists to confirm diagnosis, differentiate subtypes, or predict disease progression and treatment response.

Neurobiological mechanisms

- The interplay between the basal ganglia, cerebellum, brainstem, and cortex in the pathogenesis of dystonia is not fully elucidated.
- Dopaminergic dysfunction is implicated, but its precise role remains unclear.
- The influence of perinatal brain injury on the development of sensorimotor circuits and dystonia risk later in life is not well defined.
- The potential for early intervention to prevent or ameliorate the maladaptive development of neuronal circuits in dystonia is unexplored.

Genetic architecture

- Many genetic variants are linked to various forms of dystonia, yet the functional consequences for many genes are not fully understood.
- The global genetic landscape remains incomplete, and genotype–phenotype correlations are often ambiguous.

Research priorities

Phenotyping and biomarker discovery

- Clarify genotype–phenotype relationships to better explain clinical diversity.
- Develop biomarkers to support diagnosis, guide treatment decisions, and inform prognosis.
- Systematically characterise non-motor symptoms and patient-reported outcomes.

Genetic and molecular studies

- Implement functional studies of dystonia-associated genetic variants.
- Leverage long-read sequencing and other advanced genetic technologies to uncover additional genetic causes of disease (eg, structural and non-coding variants).

Neurodevelopmental trajectories and brain network dysfunction

- Investigate how perinatal brain injury affects the maturation of sensorimotor circuits.
- Explore crucial windows for early intervention to prevent or mitigate maladaptive network development.
- Elucidate large-scale network alterations underlying dystonia symptoms and progression.

Environmental and epigenetic influences

- Examine environmental exposures and epigenetic mechanisms contributing to disease onset and phenotypic variability.

Search strategy and selection criteria

We searched PubMed for articles published in English from Jan 1, 2020, to Aug 30, 2025, using the following terms: dystonia AND ("diagnosis" OR "classification" OR "clinical features" OR "clinical manifestations" OR "pathogenesis" OR "pathophysiology" OR "biological pathways" OR "development"). The final reference list was generated based on the relevance to the topics covered in this Review.

Conclusions and future directions

The diagnosis of dystonia is still based on clinical criteria, whereas reliable biomarkers are not available for most forms of dystonia. Research is ongoing to refine the clinical characterisation of dystonia and its phenotypical spectrum, including associated non-motor manifestations. Major advances in the field of genetics have markedly improved the aetiological diagnosis. However, unsolved issues remain regarding the interpretation of genetic variants, their penetrance, and the availability of genetic testing. Furthermore, many causes of dystonia are non-genetic. The concept of dystonia arising from alterations of sensorimotor circuits during crucial windows of development and the identification of common mechanisms among the many potential causes of dystonia indicate novel targets for therapeutic development, including the possible relevance of early intervention. We summarise knowledge gaps and research priorities in panel 2.

The new perspective of personalised therapeutic approaches raises important questions about the need for health equity to ensure that all individuals with dystonia can gain access to healthcare services with adequate genetic testing and other cutting-edge diagnostic tools, thereby avoiding missed treatment opportunities. Upcoming preclinical and translational research can target both common mechanisms and gene-specific or variant-specific effects. Another emerging field of investigation is the analysis of changes in neurophysiological activity across the dystonia network. To meet the challenge of the highly complex mechanisms involved in the pathogenesis of dystonia, future clinical-genetic studies should be based on well characterised multicentre cohorts and include multi-omics analyses and multimodal neurophysiology or neuroimaging approaches.

Contributors

ER, MV, and AP conceptualised this Review. All authors contributed to the initial literature searches; created the panels, tables, and supplementary material; wrote the first draft; and revised and edited the manuscript. All authors had final responsibility for the decision to submit for publication.

Declaration of interests

ER reports receipt of honorarium for lectures from Orkyn, Aguetant, Elivie, Merz-Pharma, Janssen, and Teva; honorarium for participating in advisory boards from Merz-Pharma, Elivie, Teva, and Bial; and research support from Merz-Pharma, Orkyn, Elivie, Everpharma, AMADYS,

ADCY5.org, Fonds de dotation Patrick Brou de Laurière, Agence Nationale de la Recherche, Dystonia Medical Research Foundation, Hope for Annabel, Cure Alternating Hemiplegia of Childhood Alternating Hemiplegia of Childhood Foundation, Alternating Hemiplegia of Childhood Association of Iceland Association française de l'hémiplégie alternante, and Alternating Hemiplegia of Childhood Kids of Netherlands. VMM is supported by a Clinician Scientist Fellowship from the Medical Research Council UK. VMM also reports receipt of research funding from the Medical Research Council UK (MRP0068681; MRP0068681) the Rosetrees Trust (CF-2021-2-112; A1598), and the Academy of Medical Sciences (Starter Grant for Clinical Lecturers); support for travel from the International Federation of Clinical Neurophysiology and the European Paediatric Neurology Society; and an honorarium for a lecture from Medtronic. LMI has received support from the Bachmann-Strauss Dystonia and Parkinson Foundation for a fellowship on dystonia research. HAJ reports receipt of grant support (recent, active, or pending) from the US Government (National Institutes of Health), private philanthropic organisations (Cure Dystonia Now and Lesch-Nyhan Syndrome Children's Research Foundation), and industry (Abbvie, Addex, Aeon, Sage, Ipsen, and Jazz). HAJ also reports serving on advisory boards or as a consultant for Abbvie, Addex, Ipsen, Merz, and Vima; receipt of stipends for administrative work from the International Parkinson's Disease and Movement Disorders Society; and serving on the scientific advisory boards for the following private foundations: the Benign Essential Blepharospasm Research Foundation, the Dystonia Medical Research Foundation, and the Tourette Association of America. HAJ is also principle investigator for the Dystonia Coalition, which has received the majority of its support through the National Institutes of Health (grants NS116025, NS065701 from the National Institutes of Neurological Disorders and Stroke, and TR001456 from the Office of Rare Diseases Research at the National Center for Advancing Translational Sciences). AP reports receipt of salary paid by the University of Pavia and IRCCS Mondino Foundation; serving as a consultant for Chiesi Farmaceutici and Idorsia pharmaceuticals; receipt of honorarium for lectures from Abbvie; and receipt of research support from Italian Ministry of Health, Italian Ministry of University, Regione Lombardia, and Dystonia Medical Research Foundation. MV declares no competing interests.

Acknowledgments

No dedicated funding was received for this Review.

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